

Pascal's triangle



- 1 The 4th row of Pascal's triangle consists of the numbers 1, 4, 6, 4, 1.
 - a Write down the numbers in the 5th row of Pascal's triangle.
 - b Find the sum of the entries in the 5th row of Pascal's triangle.
 - c Rewrite each element of the 5th row of Pascal's triangle as an nC_r coefficient.
- 2 Complete the following entries in a particular row of Pascal's triangle:
$$1, 8, \mathbb{I}, 56, 70, \mathbb{I}, 28, \mathbb{I}, 1$$
- 3 Which row of Pascal's triangle would give us the coefficients of terms in the expansion of $(x + y)^7$?
- 4 State the number of terms in the following expansions:
 - a $(4x + y)^6$
 - b $(m + y)^8$
- 5 Consider the expansion of $(y + k)^4$.
 - a How many terms will there be in the expansion?
 - b Write the coefficient of the 2nd term.
- 6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following expansion:

$$(x + m)^4 = \mathbb{I}x^4m^0 + \mathbb{I}x^3m^1 + \mathbb{I}x^2m^2 + \mathbb{I}x^1m^3 + \mathbb{I}x^0m^4$$

Binomial theorem

- 7 Evaluate the following:
 - a $\binom{6}{5}$
 - b $\binom{14}{1}$
 - c $\binom{10}{9}$
 - d $\binom{8}{0}$

8 Find the specified term for the following expansions:



a Fifth term of $(b + 2)^7$.

b Fifth term of $(c + d)^9$.

c Fifth term of $(3u - v)^6$.

d Tenth term of $\left(u + \frac{1}{2}\right)^{12}$.

e Seventh term of $\left(\frac{3y}{2} - \frac{2}{3y}\right)^{10}$.

f Eleventh term of $(2x + y^2)^{13}$.

g Sixteenth term of $(x - y^5)^{19}$.

h Middle term of $(4x^2 + 3y^3)^8$.

i Middle term of $(-3x^{-3} + 2y^{-2})^4$.

j Constant term of $\left(2y - \frac{1}{y^3}\right)^8$.

9 The first three terms of $(x + y)^{10}$ are x^{10} , $10x^9y$ and $45x^8y^2$. Using the symmetry of the coefficients, write the last three terms of the expansion.

10 In the expansion of $(m + k)^n$, the coefficient of the 9th term is 45.

a Find the value of n .

b Find the 9th term.

11 A particular term in the expansion of $\left(3a^2 + \frac{p}{b}\right)^4$ is $\frac{96a^2}{b^3}$, for some constant p . Find the value of p .

12 Consider the expansion of $(a - 4)^5$. Will the 2nd term be positive or negative?

13 By considering $(x + y)^4 = (x + y)(x + y)(x + y)(x + y)$, in how many ways can three factors of x and one factor of y be chosen from the expansion to form the term which contains x^3y^1 ? Write your answer in the form nC_r .

14 Complete the following expansion:

$$(4p + 3q)^3 = {}^3C_0 (4p)^{\text{I}} (3q)^0 + {}^3C_1 (4p)^{\text{I}} (3q)^1 + {}^3C_2 (4p)^{\text{I}} (3q)^2 + {}^3C_3 (4p)^{\text{I}} (3q)^3$$

15 Consider the binomial $(4x + 3y)^4$.

a State the first term in the expansion.

b State the last term in the expansion.

16 Consider the expansion of $(a + b)^5$.

a State the coefficient of the 3rd term in the form $\binom{n}{r}$.

b Evaluate the coefficient of the 3rd term.

c By considering the symmetry of $\binom{n}{r}$, which other term has the same coefficient as the 3rd term?

17 Which two terms in the expansion of $(u + \sqrt[3]{x})^{11}$ have a coefficient of $\binom{11}{9}$?

18 Find the coefficient of x^{17} in the expansion of $(3x^2 + 2x)^{11}$.

19 Use the binomial theorem to expand the following expressions:

a $(1 + x)^4$

b $(2 + b)^5$

c $(p + q)^5$

d $(c - d)^8$

e $(u + 4v)^6$

f $(y + 3)^4$

g $(ka - b)^6$

h $(y - 5)^3$

i $\left(y - \frac{1}{3}\right)^3$

j $\left(2 + \frac{y}{2}\right)^3$

k $(3y + 2r)^3$

l $\left(a - \frac{1}{a}\right)^3$

m $(u^2 + 3v^2)^3$

n $\left(1 + \frac{4}{y}\right)^3$

20 Find:

a The term in $(3 + \sqrt{x})^{10}$ that contains x^4 .

b The term in $(c + d)^{10}$ that contains c^3 .

c The term in $(u + v)^{11}$ that contains v^8 .

d The term in $(u + v)^{15}$ that contains u^5 .

21 If the eighth and tenth terms in the expansion of $(x + y)^n$ have the same coefficients, find the value of n .

22 Find the coefficient of a^3b^6 in the expansion of $\left(2a - \frac{b}{2}\right)^9$.

23 Consider the expansion of $(1 + x)^9$. State the coefficient of the 4th term in the form $\binom{n}{r}$.

24 Find the value of n such that the expansion of $(a + b)^n$ contains the term $84a^6b^3$.

25 Complete the following expansion by determining the missing power and binomial coefficient of the $(k + 1)$ th term:

$$(3 - x)^8 = {}^8C_0 \times 3^8 (-x)^0 + {}^8C_1 \times 3^7 (-x)^1 + \dots + \text{I} \times 3^{8-k} (-x)^{\text{I}} + \dots + {}^8C_8 \times 3^0 (-x)^8$$

26 Consider the binomial series:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

a Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.

b Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.

c Use the first five terms to approximate $\frac{1}{(1.06)^2}$ to the nearest thousandth.

